

Delhivery Network Operations Strategy Memo

Objective

Identify network bottlenecks, reduce SLA breaches, and improve ETA reliability using graph-based logistics intelligence.

Key Findings

Analysis covered more than 143,000 shipments across 1,600+ hubs. Twenty chronic delay corridors were identified. OSRM estimates significantly underperform graph-enhanced ETA models. Carting routes exhibit slightly higher delay ratios than FTL routes.

Highest Priority Bottleneck Hubs

1. IND000000ACB
2. IND562132AAA
3. IND501359AAE
4. IND212402AAA
5. IND785634AAA

Recommended Actions

Capacity Expansion: Increase sorting capacity, loading bays, and staffing at top bottleneck hubs.

Corridor Diversification: Introduce alternate routing and dynamic load balancing for chronic corridors.

Route-Type Optimization: Prefer FTL for high-volume and priority corridors; use Carting for cost-sensitive lanes.

Predictive ETA Deployment: Deploy GraphSAGE-assisted ETA prediction for planning and customer communication.

Model Performance Summary

OSRM Within $\pm 15\%$: 4.4%

Baseline ML: 50.1%

Node2Vec: 57.6%

GraphSAGE: 59.6%

Revenue-at-Risk Assessment

The analysis identified approximately **₹2.45 million** in revenue exposure associated with SLA breaches, bottleneck hubs, and chronic delay corridors.

Improvement Scenario

Assuming operational upgrades at the top three bottleneck hubs, estimated outcomes include a 10–15% reduction in corridor congestion and a 5–8% reduction in SLA breaches.

Conclusion

A small number of hubs and corridors account for a disproportionate share of operational risk. Focused investment in these locations, combined with graph-based ETA prediction and routing optimization, can materially improve network performance.